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# Comparison of logistic regression analysis with random forest for prediction of ICH early mortality in the population-based Brest Stroke Registry

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# Comparison of logistic regression analysis with random forest for prediction of ICH early mortality in the population-based Brest Stroke Registry

Thèse d'exercice de Victor Quéré[†]

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## Contents

# 1 Introduction

Intracerebral Hemorrhage (ICH) is a major cause of death and disability across the world. ICH incidence is estimated at 24.6 per 100,000 person-years, 40% of which are fatal, particularly in the acute phase with 50% of death occurring in the first two days. After such an event, it is estimated that 12 to 39% of patients regain autonomy [1]. There is currently no treatment that effectively reduces mortality after Spontaneous Intracerebral Hemorrhage (SICH) [2, 3], including intensive blood pressure management or surgery [4, 5]. In this context, it appears necessary to develop prognostic tools both to guide the decisions of clinicians and to promote the emergence of new medical or surgical therapeutic studies by identifying homogeneous group with similar prognosis.

The first risk scores were derived from logistic regression analysis, the most significant being the ICH score [6] which uses the Glasgow score (GCS), the volume of the hematoma, the presence of an Intraventricular Hemorrhage (IVH), the infratentorial origin of ICH and the patient's age to determine prognosis. However, it lacks the high degree of accuracy necessary to be used in everyday practice. This is the reason why, starting from the end of the 1990s, there was a rise of prognostic tools using artificial intelligence algorithms [7]. Since this first study, there has been a growing interest in the application of machine learning algorithms to predict SICH outcome at the initial phase. There are already seven studies on the subject which, when the comparison was carried-out, concluded that these innovative methods were superior to logistic regression equations [8–13]. However, as can be seen in the latest meta-analysis on this subject [14], superiority of these methods has not been yet proven, and there are still many areas of concern. Indeed, works are often missing some important model performance measurements, and there are methodological deficiencies regarding internal validation procedures. On the other hand, despite the existence of methods to deal with missing data, these previous studies did not report how missing variables were handled or decided to conduct a complete case-analysis. Furthermore, the variety of machine learning models compared makes it difficult to conclude on superiority and, for most of them, we deplore the lack of transparency of the algorithmic steps leading to the decision.

Thus, the purpose of this study was to develop and compare accuracy of prognostics tools regarding SICH vital prognosis at 30 days, taking into account the aforementioned findings. We trained regression-based models and Random Forest (RF) classifiers on the Brest Stroke Registry database which is one of the biggest in Europe on the subject [15]. We have chosen the RF algorithm because it requires few parameters and benefits from a good interpretability. Additionally, it already demonstrated very good performance in determining the prognosis of SICH [10].

# 2 Materials and methods

## 2.1 Data

### 2.1.1 Population

The data comes from the patient cohort of the population-based Brest stroke registry [15]. It is a prospective and multicentric database which began in 2008 and covers a population around 360,000 inhabitants of the Brest county. This area consists of 79 districts including the city of Brest and its surroundings. In order to be as exhaustive as possible, multiple data sources were selected: the emergency departments and hospital computer database of three hospitals (Brest university hospital, Brest Veteran hospital and Landerneau general hospital), regional health agency (ARS) for death certificates, general practitioners, private-practice neurologists and brain imaging recordings of the three above-mentioned hospitals or private-practice radiologists.

Adult patients with SICH, as defined by the Oxford Community Stroke Project [16] and confirmed by brain imaging between January 1, 2008 and December 31, 2017 were included. Stroke definition was verified for each case by a validation committee made up of neurologists specialized in vascular neurology. The exclusion criteria were ICH of traumatic origin, due to an arteriovenous malformation, an hemorrhagic transformation of a cerebral infarct, a cavernoma, a tumor, an intracerebral aneurysm or a cerebral venous thrombosis. Were also excluded ICH a in a post-surgical context.

### 2.1.2 Features

The target variable was death at 30 days from ICH onset. Additionally, 25 other variables were available at the time of the patient's initial assessment. We analyzed:

- **Demographic and past medical history data**: age, sex, prior Rankin score, high blood pressure, dyslipidemia, diabetes, atherosclerotic diseases (peripheral artery disease, carotid endarterectomy, coronary artery disease), cardiac arrhythmia;
- **Current drugs**: antiplatelets, anticoagulants (anti-vitamin K, heparin and direct oral anticoagulants), antihypertensive medications, paracetamol, statin and fibrates;
- **Clinical and biological data at admission**: blood glucose level, NIHSS score, GCS, heart rate, oxygen saturation, temperature, lacunar syndrome, systolic and diastolic blood pressure;

- **Imagery data**: volume and stroke localization (lobar versus non lobar).

The ICH volume was determined during the first imaging performed using the $\frac{ABC}{2}$ method, $A$ being the largest diameter of the hematoma, $B$ the diameter orthogonal to $A$ and $C$ the number of axial slices multiplied by the thickness of each cut [17]. Stroke localization was entered based on the radiologist's reviewing report. Medical history, treatments and demographic data were defined by patient self-reports.

## 2.2 Models

Each step of the modeling has been summarized on **Figure 1** and are described in the following sections. Since some modeling choices are based on the results of the descriptive analysis, the modeling choices are justified in the discussion (**Section 4**).

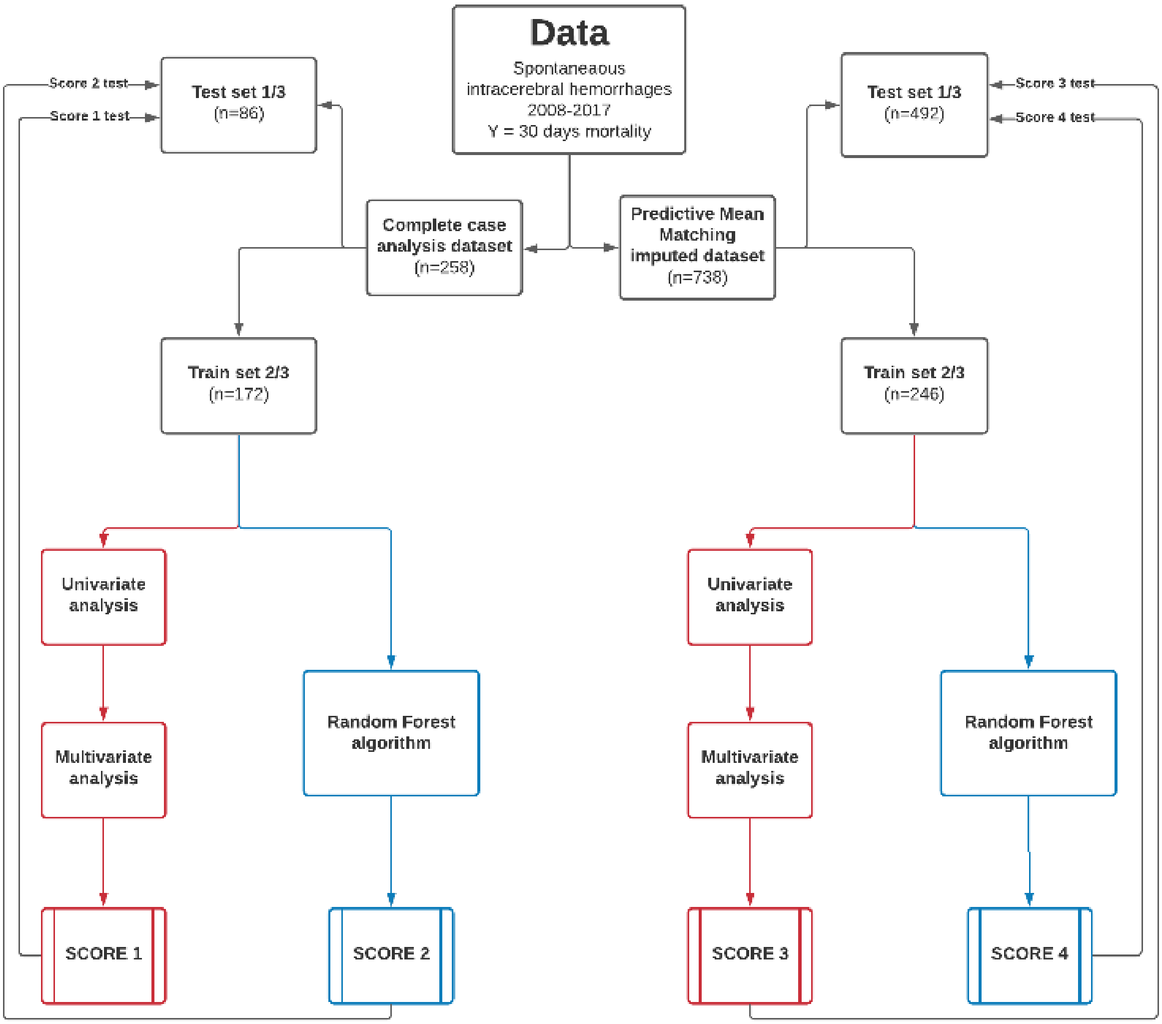



Figure 1: Description of each step of the methodology: from the complete registry, two datasets are built: 1) a complete case, gathering all subjects without a single missing piece of data; and 2) a dataset containing all subjects, complemented with a preprocessing step to impute missing entries. Each of these datasets is randomly split into 2/3 training data and 1/3 test data. Various models are then trained, and their performances are reported on the associated test sets.

### 2.2.1 Imputation

We assumed that the data was missing at random which means that missingness could be accounted for by variables where there was complete information. However, since it is impossible to verify this hypothesis without measuring the missing data, we decided to create two datasets distinguishing two cases:

- **Complete case-analysis**: In this first dataset patients with missing values were excluded;

- **Predictive Mean Matching**: In this second dataset, we included patients with and without missing data. Then, Predictive Mean Matching imputation was implemented for patients with missing predictors. This algorithm generates a regression model for each variable to be imputed. The models created make it possible to estimate a value of the variable to be imputed for each individual. The imputation is then carried out from the observed value with the closest estimate. In order to promote the good interpretability of the imputation method, we did only one iteration.

Finally, both datasets were uniform randomly split into 2/3 of data for training, and 1/3 for testing.

### 2.2.2 Statistic model

The statistical analyses were performed with SAS (Statistical Analysis System) software, version 9.4.

In the training datasets, overall frequencies or mean values were compared by Chi2 or a Fisher exact test for qualitative variables and a Student's $t$ test or Wilcoxon rank sum test for quantitative variables (**Appendix A** and **Appendix B**).

The multivariate model was realized by selecting variables significantly associated with mortality during the univariate analysis ($p < 0.20$ in the smaller dataset and $p < 0.05$ in the biggest), some confounding factors could be included without meeting these conditions. A step-down selection was applied excluding variables not contributing to the model ($p < 0.05$). Interactions between variables were not tested.

Subsequently, to implement a weighted risk score, the coefficient of the regression for each variable was divided by the smallest coefficient and rounded to the nearest integer to derive points to associate with individual variables [18]. Next, identified scores were applied to the training datasets to select the better scores thresholds thanks to the Younden J statistic [19]. We finally tested those scores on the test datasets to compute AUC, sensitivity (Se), specificity (Sp), positive predictive values (PPV) and negative predicted values (NPV). The goal was to maximize the area under the curve (AUC).

### 2.2.3 Random Forest algorithm

Machine learning is a field of artificial intelligence which is based on mathematical and statistical approaches to allow computers to solve tasks without being explicitly programmed to do so. The creation of a machine learning model consists of two steps. The first step, called training phase, estimates a model to solve a problem from a sample of the data, called training data. When the expected result of the problem is accessible during this step, the learning is said to be supervised. Moreover, we speak about classification when the result is a discrete variable and about regression when it is a continuous variable. During the second step, called testing phase, we submit new data (unavailable during training), called testing data, to the created model to estimate its predictive performance. A common practice to train better models is to use a validation set. To do so, the training dataset is randomly split into actual training data, and validation data. Models are then trained on the former, and evaluated on the latter. While this gives access to less data during training, it encourages a better generalization on unseen data, making trained models more robust. One of the most advanced methods is Cross Validation (CV). This amounts to randomly dividing the training data into $k$ samples. One of the samples is designed as the validation set and the rest as the training set. The performance of the model, created on the learning set, will thus be able to be evaluated on the validation set. The operation will be repeated until each of the $k$ samples has been used as a validation set. In the end, we obtain $k$ performance scores, the mean and standard deviation of which represent an overall score over the $k$ validation sets.

In this work, we used a RF algorithm to classify patients according to their 30-day mortality. RF are supervised machine learning algorithms used for classification and regression. They belong to the category of ensemble learning methods by using multiple Decision Trees (DT) to make predictions. DT are binary trees dividing a population of individuals into homogeneous groups according to a set of predictors and to a target variable. At each internal node, the DT selects the predictor which maximizes homogeneity of the subsets. Thus, each branch represents the outcome of the selection and each leaf node is a class label. The trees then grow until they meet the conditions defined in their parameters or if they manage to create only homogeneous groups. In our case, the chosen criteria for selecting predictors at each node was Gini Impurity (GI). GI is the probability of a new individual being incorrectly classified at a given node if it were randomly labeled according to class distribution in the dataset. The GI thus estimates the impurity of created subsets at each node, a GI of zero reflecting a minimum impurity and a GI of one a maximum impurity.

DT have many qualities such as their ability to work with variables of all types without distribution assumptions, their simplicity of interpretation or their easy visualization which make them intuitive models. Unfortunately, they are also known for their instability which means that small variations in the data can lead to very different tree architectures. Moreover, this algorithm suffers from a significant propensity to overfitting, which is too much fidelity to training data, resulting in poor generalization and therefore an alteration of the predictive powers of the model [20].

It is by combining random subspace method [21] and bootstrap aggregating [22] that RF manages to partially correct these defects. Random subspace method attempts to create diversity among decision trees by training them on random samples of predictors. In the same way, bootstrap aggregating generates new training sets by sampling observations. It is also the architecture itself of the RF which reduces overfitting and increases stability. For example, in classification problems like mortality, it is the majority vote of the DT that makes it possible to determine the class. Although it is criticized to be a "black box" like most algorithms, weights of predictors regarding the classification can be estimated by some metrics like Mean Decrease in Gini (MDG). Since RF are an ensemble of DT, we can calculate MDG which is the average of decrease in node impurity for this variable weighted by the proportion of samples reaching that node in each individual tree of the forest. This metric was calculated for both machine learning algorithms.

The `RandomForestClassifier` function from the scikit-learn Python library, version 0.24.2 [23] was used. The code was written in Python and is freely accessible[1].

The different parameters tested were: the number of trees in the forest ($n\_estimators$), the maximum depth of trees ($max\_depth$), the minimum number of samples required to split an internal node ($min\_samples\_split$), and the number of features to consider when looking for the best split ($max\_features$). In order to select the best hyperparameters, we used the `GridSearchCV` function from the scikit-learn library, which makes it possible to test alternately all the combinations of parameters considered. It integrates a 5-fold cross-validation to reduce overfitting. The goal was to maximize the AUC.

We finally tested those scores on the test datasets to calculate AUC, Se, Sp, PPV and NPV.

# 3 Results

## 3.1 Description of populations

We collected data from 738 SICH between 2008 and 2017. Considering the dataset before imputation, the average age was 73.2 years old and 45.7% were male. 24.1% were treated with antiplatelets agents and 20.7% with anticoagulants. The overall 30-day mortality was 33.3%. All other characteristics are summarized in **Appendix C**.

Regarding the missing data, only history of Rankin score (27.7%), body temperature

[1] `https://github.com/VictorQuere/Method-comparison-to-evaluate-the-prognosis-of-SICH`

(29.0%) and blood glucose level (32.0%) had more than 20% of missing observations. The variables like entry treatment (antihypertensive, antiplatelet, anticoagulant, statin, fibrate and paracetamol), history of atherosclerotic diseases, age, sex had no missing values. It was also quite clear that there was more missing data for patients who died. Particularly for the history of Rankin score (15.2% versus 52.9%), the location of the hemorrhage (0.0% versus 12.2%) and the volume of hemorrhage (4.1% 18.3%).

Regarding the imputed dataset, thanks to the imputation method which uses variables previously observed, the metrics seemed not to differ significantly from the original non imputed dataset, as shown in **Appendix D**. However, although the statistical comparison was not made, the complete case-analysis dataset looked to be different from the imputed dataset as given in **Appendix E**. Indeed, the overall 30-day mortality was 13.5% in complete dataset versus 33% in original dataset. Furthermore, there seemed to be a lower proportion of patients without missing data with a GCS lower than 5 (11.1% versus 3.1%) and more patients with a GCS above twelve (79.8% versus 66.0%). Similarly, there was a lower percentage of patients with an NIHSS greater than 15 (25.2% versus 35%) and a volume greater than 30 (27.9% versus 35.7%) in the complete dataset.

## 3.2 Models development

### 3.2.1 Statistical models development

Continuous variables were dichotomized for ease of use of the score with thresholds in line with previous studies. For blood glucose a threshold >1.2 g/L was arbitrarily set [24], it was >37,5°C for body temperature [25] and >30mL for volume of hemorrhage [26, 27]. In the same way, discrete variables were divided into subcategory: GCS: 3 to 4, 5 to 12, and 13 to 15) [28], NIHSS (0 to 5, 6 to 15 and >15) [29, 30] and history of Rankin score (0,1,2 or >2). For the age we defined a threshold >75 year-old which corresponded to the definition of old age.

#### Complete case-analysis dataset

A step-down selection was applied excluding variables not contributing to the model ($p < 0.05$). Age $\geq$ 75 years ($OR = 3.77$, $p = 0.0267$), history or discovery of cardiac arrhythmia ($OR = 4.09$, $p = 0.0158$), GCS $< 5$ ($OR = 25.08$, $p = 0.0007$), GCS between 5-12 ($OR = 6.77$, $p = 0.0007$) and hemorrhage volume $\geq 30$ ($OR = 3.57$, $p = 0.0229$) were identified as significant mortality predictors on the training dataset. For each variable, regression coefficient was divided by the smallest coefficient (1.27 from hemorrhage volume in this case) and rounded to the nearest integer to derive the individual variable weight.

1 point was awarded if the patient age was greater or equal to 75 years, 1 point if there was history or discovery of cardiac arrhythmia, 2 points if the GCS was strictly less than 5, 1 point if the GCS was between 5-12 and 1 point if the hemorrhage volume was greater or equal to 30cm$^3$. The final score had a theoretical value between 0 and 5 (**Table 1**).

| | | OR [IC95%] | p-value | Regression coeficient | Score Points |
|---|---|---|---|---|---|
| Age | < 75 y | 1.00 (réf) | 0.0267 | | 0 |
| | >=75 y | 3.77 [1.17 ; 12.18] | | 1.3263 | 1 |
| Cardiac arrhythmia | No | 1.00 (réf) | 0.0158 | | 0 |
| | Yes | 4.09 [1.30 ; 12.82] | | 1.4075 | 1 |
| Glasgow Score | <5 | 25.08 [2.80 ; 224.35] | 0.0007 | 3.2220 | 2 |
| | 5-12 | 6.77 [2.13 ; 21.53] | | 1.9120 | 1 |
| | >12 | 1.00 (réf) | | | 0 |
| Hemorrhage volume (cm³) | < 30 | 1.00 (réf) | 0.0229 | | 0 |
| | >=30 | 3.57 [1.19 ; 10.69] | | 1.2731 | 1 |
| Total score | | | | | 0-5 |

Arrhythmia could be discovered during the initial presentation or be an history of cardiac arrhythmia. Glasgow score was calculated on initial presentation. Hemorrhage volume was determined on initial CT calculated using ABC/2 method.

Table 1: **Score 1**, based on complete-case analysis

### Predictive Mean Matching imputed dataset

A step-down selection was applied excluding variables not contributing to the model ($p < 0,05$). Age $\geq$ 75 years ($OR = 1.81$, $p = 0.0382$), history of atherosclerotic disease ($OR = 2.34$, $p = 0.0085$), blood glucose level $>$ 1.2g/L ($OR = 2.60$, $p = 0.0010$), lacunar syndrome ($OR = 6.15$, $p = 0.0227$), GCS $<$ 5 ($OR = 58.01$, $p < 0.001$), GCS between 5-12 ($OR = 4.81$, $p < 0.001$), absence of lobar hemorrhage ($OR = 5.31$, $p \leq 0.001$), hemorrhage volume $\geq$30 ($OR = 7.18$, $p < 0.001$) and prior anticoagulant medication ($OR = 2.67$, $p = 0.0022$) were identified as significant mortality predictors on the training dataset. For each variable, regression coefficient was divided by the smallest coefficient (0.5960 from age in this case) and rounded to the nearest integer to derive the individual variable weight. 1 point was awarded if the patient age was greater than or equal to 75 years, 1 point if there was an history of atherosclerotic disease, 2 points if blood glucose level was strictly greater than 1.2g/L on the first blood test, 3 points if there was a lacunar syndrome on initial presentation, 7 points if GCS was $<$ 5 , 3 points if GCS was between

5-12, 3 points if there was an absence of lobar hemorrhage, 3 points if the hemorrhage volume was greater or equal to $30cm^3$ and 2 points if there was prior anticoagulant medication. The final score had a theoretical value between 0 and 25 (**Table 2**).

| | | OR [IC95%] | p-value | Regression coeficient | Score points |
|---|---|---|---|---|---|
| Age | < 75 y | 1.00 (réf) | 0.0382 | | 0 |
| | >=75 y | 1.81 [1.03 ; 3.20] | | 0.5960 | 1 |
| Artheromatic disease | No | 1.00 (réf) | 0.0085 | | 0 |
| | Yes | 2.34 [1.24 ; 4.40] | | 0.8488 | 1 |
| Blood glucose level (g/L) | <=1.2 | 1.00 (réf) | 0.0010 | | 0 |
| | >1.2 | 2.60 [1.47 ; 4.60] | | 0.9561 | 2 |
| Syndrome lacunaire | No | 6.15 [1.30 ; 29.31] | | 1.8158 | 3 |
| | Yes | 1.00 (réf) | 0.0227 | | 0 |
| Glasgow score | <5 | 58.01 [17.19 ; 195.75] | | 4.0607 | 7 |
| | 5-12 | 4.81 [2.73 ; 8.46] | | 1.5702 | 3 |
| | >12 | 1.00 (réf) | <0.001 | | 0 |
| Lobar hemorrhage | No | 5.31 [2.76 ; 10.23] | | 1.6700 | 3 |
| | Yes | 1.00 (réf) | <0.001 | | 0 |
| Hemorrhage volume ($cm^3$) | < 30 | 1.00 (réf) | <0.001 | | 0 |
| | >=30 | 7.18 [3.79 ; 13.61] | | 1.9713 | 3 |
| Prior anticoagulant medication | No | 1.00 (réf) | 0.0022 | | 0 |
| | Yes | 2.67 [1.42 ; 5.01] | | 0.9829 | 2 |
| Total score | | | | | 0-25 |

History of artheromatic diseases could be peripheral arterial obstructive disease, operated carotid stenosis or coronary artery disease. Blood glucose level was determined on the first blood test at emergency room. Glasgow score was calculated on initial presentation. Imagery data were determined on initial CT and volume was calculated using ABC/2 method. Prior anticoagulation medication included anti-vitamin K, heparin and direct oral anticoagulants.

Table 2: **Score 3**, based on imputed dataset

### 3.2.2 Random Forest algorithm development

RF were trained without prior variable selection on both training datasets. The reason being that these algorithms can carry out a random selection of a portion of the variables at each node of the decision trees in the forest. In order to save computational time, we analyzed a validation curve of number of estimators. Validation curve is a graph which represents training scores and validation scores according to a determined parameter. This curve makes it possible to define the best adjustment of a hyperparameter which results in a good performance without overfitting. In our case, by varying the number of estimators ($n\_estimators$) between 100 and 2000, we saw that the performance gain was negligible. We therefore decided to set this number to 100 which requires less resources to perform calculations (**Appendix G**). Then, to find the most robust parameters, we tested multiple parameters thanks to a gridsearch algorithm with a 5-fold cross-validation. As mentioned in (**Section 2.2.3**), we chose to use Gini Impurity as a measure of the homogeneity of the labels at each node and we authorized bootstrapping in order to reduce the risk of overfitting. All the tested parameters are summarized in **Appendix F**.

**Complete case-analysis dataset**

The best parameters were: no max depth in the decision trees, no maximum number of feature to consider at each split, min samples split of 47 and 100 estimators. All variables were used by the RF, the three variables standing out from the others being GCS, volume of hemorrhage and NIHSS score. The complete list of features and associated importance is shown in **Score 2** in **Figure 2**.

**Predictive Mean Matching imputed dataset**

The best parameters were: no max depth in the decision tress, a maximum number of feature to consider at each split of $log2(n\_estimators)$, min samples split of 10 and 100 estimators. Prior antiplatelet, paracetamol, antihypertensive, or anticoagulant medication were not used by the RF. In the same way, history of dyslipidemia, history of high blood pressure, history of diabetes, hemorrhage localisation and lacunar syndrome did not contribute to classifications. The five variables standing out from the others were NIHSS score, volume of hemorrhage, GCS, age and blood glucose level. The complete list of features and associated importance is shown in **Score 4** in **Figure 2**.

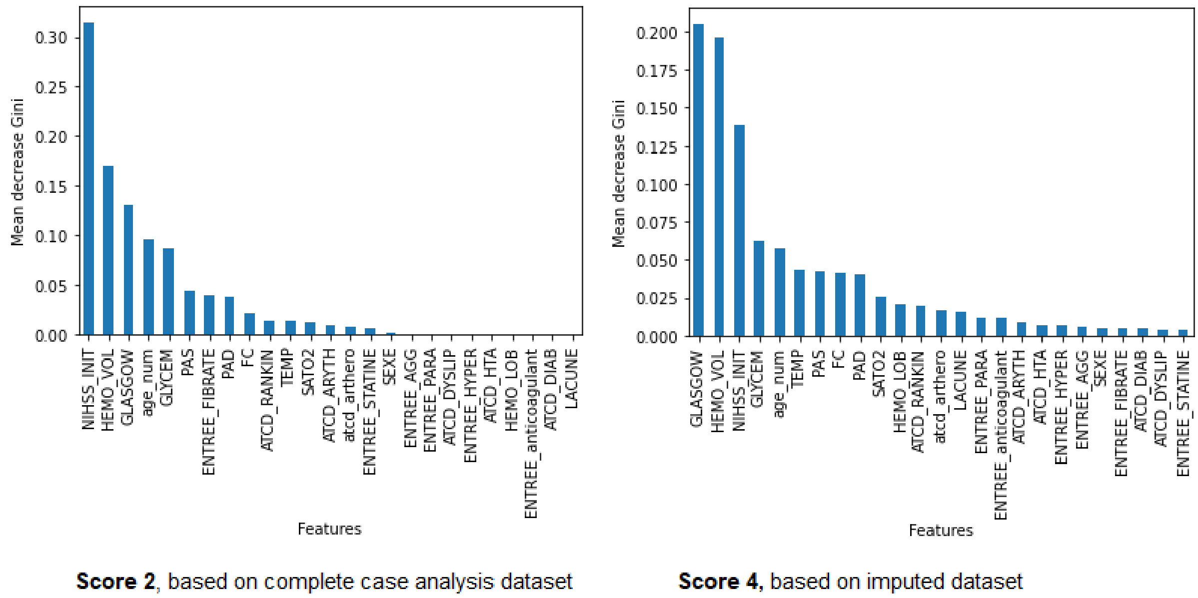



Figure 2: Feature importance of **Score 2** and **Score 4** using mean decrease Gini

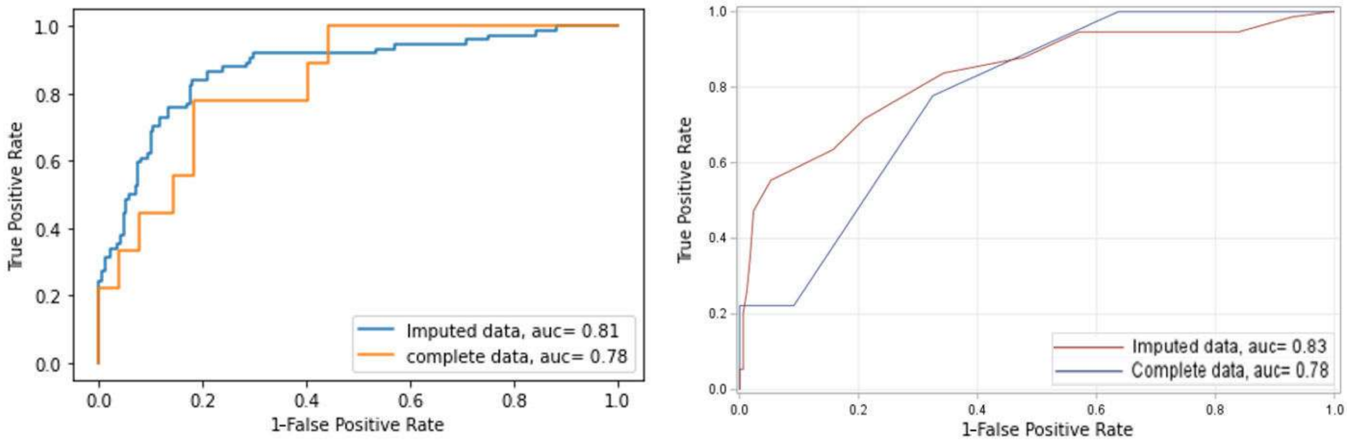
## 3.3 Models results

### 3.3.1 Results on complete case-analysis dataset

On the training complete case-analysis dataset (n=172), in which 18% of patients died, regression based model (**Table 1**) reached an AUC of 0.81 with best score threshold being 3 and RF obtained an average AUC of 0.75. On the testing complete case-analysis dataset (n=86), in which only 10% of patients died, statistic score AUC was 0.78, Se 0.22, Sp 0.91, PPV 0.22 and NPV 0.91. RF had an AUC of 0.78, Se 0.78, Sp 0.78, PPV 0.29 and NPV 0.96.

### 3.3.2 Results on Predictive Mean Matching imputed dataset

On the training imputed dataset (n=492), in which 36% of patients died, regression based model (**Table 2**) reached an AUC of 0.89 with best score threshold being 10 and RF obtained an average AUC of 0.90. On the testing complete case-analysis dataset (n=246), in which 30% of patients died, statistic score AUC was 0.83, Se 0.72, Sp 0.79, PPV 0.60 and NPV 0.87. RF had an AUC of 0.81, Se 0.76, Sp 0.87, PPV 0.71 and NPV 0.89. Comparison of Receiver Operating Characteristic (ROC) curves confirmed the best performances of **Score 3** (**Figure 3**).




**A.** ROC of RF algorithms

**B.** ROC of regression based models

Figure 3: Comparison of ROC curves of the four different scores

# 4 Discussion

Today, the creation of prognostic scores for SICH is essential both to guide the management of clinicians and to define homogeneous groups of patients which could lead to the emergence of new therapeutic studies. Beyond performance, it is sometimes the architecture of prognostic tools that allows progress, for example by providing better explainability or ease of use. Likewise, the comparison of methodologies teaches us more about the prognosis of SICH and its predictors. It is with these objectives that we developed and compared prognostic models using two distinct methodologies. We therefore first concentrated our efforts on creating a database representative of the target population, in particular thanks to imputation of missing data. Then, based on good internal validation procedures, we were able to create precise models and highlight the potential of RF as a prognostic tool.Thanks to this comparison of methods, we were able to confirm the central role of certain predictors and discover the potential of others.

However, before discussing these results, it will be necessary to take into account several limitations to our study:

- **Comparison to ICH score is missing**: Unfortunately we could not compare our models to the ICH score [6] due to the lack of data concerning the presence of intraventricular hemorrhage. This data is currently being extracted and will be integrated into our future studies, especially since it is known to be an important predictor [31, 32]. In the same way, imaging markers like spot sign, black hole sign, hypodensities and island sign, known to be independently related to the prognosis in the acute phase of ICH [33–35], were not available. Additionally, analyzing raw imaging data could have improved the accuracy of our artificial intelligence algorithms. Indeed, several studies concerning the prognosis of ischemic strokes have already shown improvements when using raw imaging data as input [36–38];

- **Withdrawal of active treatment was not taken into account**: Indeed, some patients must have had withdrawal of active treatment after severe SICH. However, as underlined by several studies [39, 40], practitioners and scores might overestimate the risk of death or severe disability after a SICH. Consequently, this could lead to a self-fulfilling prophecy by providing suboptimal care for severe patients. In order to prevent this bias from being reflected in the scores, it would have been necessary to inform the patients for whom the care was limited;

- **Other machine learning models could have been used**: Moreover, we didn't use other machine learning algorithms that could have performed better. For example, neural networks appear to be the algorithms with the greatest prognostic precision in the latest meta-analysis on the subject [14]. However, this performance gain comes at the expense of interpretability, which encouraged us to prefer RF;

- **It is possible to refine the estimated outcomes**: Finally, it would be interesting to evaluate the functional prognosis among the surviving patients. We could have used the Rankin score at 3 months to refine their outcome. This would possibly make it possible to form homogeneous groups of patients for therapeutic studies, or simply to guide clinicians in their decisions. Unfortunately, this would have reduced the size of our dataset due to the significant percentage of missing data.

Now that these limits have been presented, let us look at the promising results of this work:

- **The largest database has been analyzed**: With more than ten times the number of patients in the originator study [7], this is to date, to our knowledge, the largest database of SICH analyzed using machine learning. In addition to its size, the Brest stroke registry is exhaustive and benefits from a good external validity thanks to a prospective retrieval of data from multiple sources [15];

- **Data imputation of missing values improves the representativity of the population**: The analysis of such a large dataset was also possible thanks to the processing of missing predictors. Actually, most of the studies related to SICH prognosis did not describe how they handled missing data, or chose to exclude patients with missing observations [14]. By doing so, they could have reduced the size of their datasets and therefore the power of their statistical analyzes. In our case, removing observations with missing data reduced the sample size by 65% (258 patients versus 738) with poorer results of the models. Beyond the loss of performance, exclusion of patients with missing data may have resulted in attrition bias. Indeed, the statistical comparison of the datasets is currently in progress but preliminary analyzes seemed to show differences between them with more missing data in more severe patients. For instance, graphical analysis of heatmap of missing data based on GCS (**Appendix J**) seemed to show more missing predictors like history of Rankin score or hemorrhage volume in comatose patients. Moreover, as described in **Section 3.1**, we noticed there was a greater proportion of patients with high NIHSS (35% versus 25.2%), low GCS (3.1% versus 11.1%) and large volumes of hemorrhage (35.7% versus 27.9%) in the original dataset. In the same way the lowest percentage of deaths in the complete case-analysis (13.5% versus 33%), reinforced this assumption. This makes sense since it is more difficult to collect historical data in patients with impaired consciousness. Withdrawal of care is also more frequent in patients with a poor prognosis which limits explorations. To verify this hypothesis, it will be essential that future studies regarding SICH prognosis try to inform the reason why variables are missing while collecting them. Otherwise, imputation methods will have to be used to limit their impact. To impute quantitative and qualitative data, Predictive Mean Matching seems to be a good solution. This is an already well-tested multiple imputation method which has the advantage of imputing from observed values which reduces the risk of producing outliers [41]. Thus, apart from sometimes reinforced correlations, the imputed dataset remains fairly close to the original. This was confirmed in our case with very similar descriptive analyzes between the imputed dataset (**Appendix D**) and the original (**Appendix C**). It would finally be great to test our scores on external datasets to judge their robustness independently of the treatment of missing data applied to them;

- **Data imputation improves the prediction power of both the statistical and machine learning models**: Furthermore, the imputation of missing data has enabled us to improve the performance of our models. It is indeed the models trained on the imputed databases which obtained the best AUC, with 0.83 against 0.78 for the regression models and 0.81 against 0.78 for the RF. On the other hand, within the same dataset, whatever the method used, the results remained similar. **Score 3** (**Table 2**) obtained highest AUC of 0.83 but very close to RF with 0.81 on the same data. And, although the RF trained on the imputed data obtained better results in terms of Se, Sp, PPV and NPV, this could be related to a threshold effect. That's why the AUC remained the most reliable metric to assess the performance of prognostic or diagnostic tools. In fact, the distribution of the results of deceased or non-deceased patients has an area of overlap. Therefore, it is necessary to define a threshold value to distribute the results and any choice of threshold leads to classification errors having repercussions on the sensitivity and specificity values associated with it. The ROC curve makes it possible to represent the Se and Sp of a model for all the possible threshold values. Thus, the comparison of AUCs appears to be more robust for evaluating these tools [42];

- **Random forests provide interpretable results**: Besides prediction performance of the models, other characteristics must be discussed. Although only a little training is needed to use machine learning algorithm as prognostic tools, they still suffer from a "black box" effect, particularly criticized in healthcare when decisions can have serious consequences. It is often said that, compared to machine learning, regression models benefit from a good interpretability and ease of use by integrating fewer variables and by not requiring the use of computers. In this context, the choice to use RF seems to be relevant. Indeed, in addition to their ability to work with variables of all types without distribution assumptions and without too much hyperparameters, RF benefits from a simpler and more understandable architecture than many algorithms. It is related to the fact that an RF algorithms are made up of many Decision Trees which are easy to interpret and visualize. Moreover, due to the diversity of trees making up the forest, this makes RF naturally resilient to overfitting and therefore ensures better generalization of the results. Then, as mentioned in **Section 2.2.3**, the joint use of random subspace method and bootstrap aggregating further reduces the risk of overfitting. Also, it is possible to evaluate the weight of the variables in the decisions of the RF as we did with the MDG (**Figure 2**) what participates in the interpretability of this model. Finally, the use of such models makes it possible to employ internal validation methods such as cross validation. By alternately selecting samples of the data as a validation set, it allows better external validity and less overfitting even on smaller data sets;

- **GCS, hemorrhage volume, age, hyperglycemia on arrival, NIHSS, prior anticoagulant medication and prior fibrates medication appear to be important predictors in SICH prognosis**: The comparison of regression-based and machine learning models also makes it possible to confirm the importance of certain predictors in SICH prognosis at the acute phase. GCS, hemorrhage volume and age were already well-known predictors but other newer variables seem to confirm their importance. Despite 30% of missing data, hyperglycemia on arrival at the emergency room is decisive in all the scores. This finding is in agreement with a recent meta-analysis which collects the results of 16 studies and tends to show that the rise in blood sugar is linked to a poor prognosis at the acute phase of SICH [24]. On the other hand, NIHSS score has a great importance in classification of our RF whereas it is not discriminating in the regression models. It could be the consequence of the threshold dichotomization carried out before statistical analysis. It is also important to underline that the value of the GI tends to linearly increase with the number of values of an attribute which could have increased the weight attributed to the NIHSS in the RF [43]. Nevertheless, as pointed out by some recent studies, NIHSS score seems to have a potential interest to assess the fate of SICH regardless of the GCS [44, 45]. Independently analyzing the items of NIHSS could lead to improvement in future studies. Regarding the regression based scores, we can also notice that prior anticoagulant medication seems to play an important role in the prognosis, especially considering that the cardiac arrhythmia could be a colinear variable since most of the patients suffering from it are treated with anticoagulants. Lastly, prior fibrates medication were found to be a protective factor only in Score 2 (**Figure 2**), this is probably a specific characteristic of the population studied, but some fundamental research papers described a possible neuroprotective effect of fibrates after ICH in rats [46]. To our knowledge, no clinical study has yet reported this potential effect in humans, possibly because this treatment had never been studied in this context before.

# 5 Conclusion

The objective of this study was to develop and compare prognostic models of the risk of early mortality after SICH, taking into account the impact of missing data from an innovative methodology described in **Figure 1**. We thus compare the performance of weighted scores from regression models with RF algorithms.

The results showed better performance of the models applied on the imputed data. RF algorithms have also proven to be good candidates for developing health prognostic tools thanks to their interpretability, precision and ease of use. In addition to improving model performance, the imputation of missing data could have contributed to a better representativity of the samples studied. Furthermore, the method comparison made it possible to confirm the importance of certain variables such as GCS and the volume of hemorrhage, but also to participate in the emergence of newer predictors such as NIHSS, blood glucose level or anticoagulants.

Finally, our work opens up interesting perspectives, especially regarding the treatment of missing predictors. The underlying causes should be documented during future data collection to minimize the risk of attrition bias. In addition, the use of raw data could participate in improving the performance of artificial intelligence models. The comparison should then be made with current scores such as the ICH score to estimate the potential performance gain.

# Appendices

## A Univariate analysis of the complete case-analysis training dataset

| | | Total (N= 172) | Alive at 30 days (N= 146) | Dead at 30 days (N= 26) | P |
|---|---|---|---|---|---|
| Age | < 75 y | 79 (45.9%) | 73 (50.0%) | 6 (23.1%) | 0.0111 |
| | >=75 y | 93 (54.1%) | 73 (50.0%) | 20 (76.9%) | |
| Sex | Male | 86 (50.0%) | 71 (48.6%) | 15 (57.7%) | 0.3945 |
| | Female | 86 (50.0%) | 75 (51.4%) | 11 (42.3%) | |
| Artheromatic diseases | No | 149 (86.6%) | 129 (88.4%) | 20 (76.9%) | 0.1239 |
| | Yes | 23 (13.4%) | 17 (11.6%) | 6 (23.1%) | |
| Cardiac arrhythmia | No | 127 (73.8%) | 112 (76.7%) | 15 (57.7%) | 0.0421 |
| | Yes | 45 (26.2%) | 34 (23.3%) | 11 (42.3%) | |
| Hypertension | No | 78 (45.3%) | 67 (45.9%) | 11 (42.3%) | 0.7353 |
| | Yes | 94 (54.7%) | 79 (54.1%) | 15 (57.7%) | |
| Diabetes | No | 145 (84.3%) | 123 (84.2%) | 22 (84.6%) | 1.0000 |
| | Yes | 27 (15.7%) | 23 (15.8%) | 4 (15.4%) | |
| Dyslipidemia | No | 127 (73.8%) | 109 (74.7%) | 18 (69.2%) | 0.5619 |
| | Yesi | 45 (26.2%) | 37 (25.3%) | 8 (30.8%) | |
| Systolic blood pressure | N | 172 | 146 | 26 | 0.8527 |
| | Mean +/- SD | 171.42 +/- 31.91 | 171.53 +/- 31.83 | 170.81 +/- 33.02 | |
| Diastolic blood pressure | N | 172 | 146 | 26 | 0.9472 |
| | Mean +/- SD | 88.94 +/- 18.22 | 89.27 +/- 18.33 | 87.08 +/- 17.78 | |
| Body temperature (°C) | <=37.5 | 156 (90.7%) | 135 (92.5%) | 21 (80.8%) | 0.0713 |
| | >37.5 | 16 (9.3%) | 11 (7.5%) | 5 (19.2%) | |
| Heart rate | N | 172 | 146 | 26 | 0.6011 |
| | Mean +/- SD | 80.96 +/- 15.96 | 80.62 +/- 15.74 | 82.85 +/- 17.36 | |
| Oxygen saturation | N | 172 | 146 | 26 | 0.9485 |
| | Mean +/- SD | 96.31 +/- 3.41 | 96.31 +/- 3.45 | 96.35 +/- 3.24 | |
| Blood glucose level (g/L) | <=1.2 | 84 (48.8%) | 77 (52.7%) | 7 (26.9%) | 0.0153 |
| | >1.2 | 88 (51.2%) | 69 (47.3%) | 19 (73.1%) | |
| Lacunar syndrome | No | 153 (89.0%) | 128 (87.7%) | 25 (96.2%) | 0.3138 |
| | Yes | 19 (11.0%) | 18 (12.3%) | 1 (3.8%) | |
| History of Rankin score | 0 | 83 (48.3%) | 77 (52.7%) | 6 (23.1%) | 0.0229 |
| | 1 | 40 (23.3%) | 31 (21.2%) | 9 (34.6%) | |
| | 2 | 14 (8.1%) | 10 (6.8%) | 4 (15.4%) | |
| | >2 | 35 (20.3%) | 28 (19.2%) | 7 (26.9%) | |

| | | Total (N= 172) | Alive at 30 days (N= 146) | Dead at 30 days (N= 26) | P |
|---|---|---|---|---|---|
| Glasgow score | <5 | 5 (2.9%) | 2 (1.4%) | 3 (11.5%) | <0.0001 |
| | 5-12 | 31 (18.0%) | 19 (13.0%) | 12 (46.2%) | |
| | >12 | 136 (79.1%) | 125 (85.6%) | 11 (42.3%) | |
| NIHSS | <=5 | 78 (45.3%) | 75 (51.4%) | 3 (11.5%) | 0.0006 |
| | 6-15 | 51 (29.7%) | 40 (27.4%) | 11 (42.3%) | |
| | >15 | 43 (25.0%) | 31 (21.2%) | 12 (46.2%) | |
| Lobar hemorrhage | No | 96 (55.8%) | 82 (56.2%) | 14 (53.8%) | 0.8264 |
| | Yes | 76 (44.2%) | 64 (43.8%) | 12 (46.2%) | |
| Hemorrhage volume ($cm^3$) | < 30 | 121 (70.3%) | 110 (75.3%) | 11 (42.3%) | 0.0007 |
| | >=30 | 51 (29.7%) | 36 (24.7%) | 15 (57.7%) | |
| Prior antihypertensive medication | No | 94 (54.7%) | 80 (54.8%) | 14 (53.8%) | 0.9287 |
| | Yes | 78 (45.3%) | 66 (45.2%) | 12 (46.2%) | |
| Prior antipatelet medication | No | 129 (75.0%) | 111 (76.0%) | 18 (69.2%) | 0.4609 |
| | Yes | 43 (25.0%) | 35 (24.0%) | 8 (30.8%) | |
| Prior anticoagulation medication | No | 141 (82.0%) | 121 (82.9%) | 20 (76.9%) | 0.5789 |
| | Yes | 31 (18.0%) | 25 (17.1%) | 6 (23.1%) | |
| Prior statin medication | No | 145 (84.3%) | 124 (84.9%) | 21 (80.8%) | 0.5656 |
| | Yes | 27 (15.7%) | 22 (15.1%) | 5 (19.2%) | |
| Prior fibrate medication | No | 166 (96.5%) | 142 (97.3%) | 24 (92.3%) | 0.2245 |
| | Yes | 6 (3.5%) | 4 (2.7%) | 2 (7.7%) | |
| Prior paracetamol medication | No | 127 (73.8%) | 108 (74.0%) | 19 (73.1%) | 0.9237 |
| | Yes | 45 (26.2%) | 38 (26.0%) | 7 (26.9%) | |

# B Univariate analysis of the imputed training dataset

| | | Total (N= 492) | Alive at 30 days (N= 320) | Deat at 30 days (N= 172) | P |
|---|---|---|---|---|---|
| Age | < 75 y | 209 (42.5%) | 161 (50.3%) | 48 (27.9%) | <0.0001 |
| | >=75 y | 283 (57.5%) | 159 (49.7%) | 124 (72.1%) | |
| Sex | Male | 222 (45.1%) | 150 (46.9%) | 72 (41.9%) | 0.2865 |
| | Female | 270 (54.9%) | 170 (53.1%) | 100 (58.1%) | |
| Artheromatic diseases | No | 395 (80.3%) | 270 (84.4%) | 125 (72.7%) | 0.0019 |
| | Yes | 97 (19.7%) | 50 (15.6%) | 47 (27.3%) | |
| Cardiac arrhythmia | No | 380 (77.2%) | 263 (82.2%) | 117 (68.0%) | 0.0004 |
| | Yes | 112 (22.8%) | 57 (17.8%) | 55 (32.0%) | |
| Hypertension | No | 208 (42.3%) | 138 (43.1%) | 70 (40.7%) | 0.6033 |
| | Yes | 284 (57.7%) | 182 (56.9%) | 102 (59.3%) | |
| Diabetes | No | 441 (89.6%) | 284 (88.8%) | 157 (91.3%) | 0.3802 |
| | Yes | 51 (10.4%) | 36 (11.3%) | 15 (8.7%) | |
| Dyslipidemia | No | 366 (74.4%) | 237 (74.1%) | 129 (75.0%) | 0.8203 |
| | Yes | 126 (25.6%) | 83 (25.9%) | 43 (25.0%) | |
| Systolic blood pressure | N | 492 | 320 | 172 | 0.5306 |
| | Mean +/- SD | 169.50 +/- 33.96 | 170.98 +/- 31.67 | 166.74 +/- 37.79 | |
| Diastolic blood pressure | N | 492 | 320 | 172 | 0.6714 |
| | Mean +/- SD | 89.24 +/- 18.83 | 90.09 +/- 19.05 | 87.66 +/- 18.37 | |
| Body temperature (°C) | <=37.5 | 455 (92.5%) | 302 (94.4%) | 153 (89.0%) | 0.0297 |
| | >37.5 | 37 (7.5%) | 18 (5.6%) | 19 (11.0%) | |
| Heart rate | N | 492 | 320 | 172 | 0.8036 |
| | Mean +/- SD | 82.03 +/- 18.24 | 81.93 +/- 17.28 | 82.23 +/- 19.95 | |
| Oxygen saturation (%) | N | 492 | 320 | 172 | 0.1229 |
| | Mean +/- SD | 96.67 +/- 3.10 | 96.70 +/- 2.37 | 96.62 +/- 4.13 | |
| Blood glucose level (g/L) | <=1.2 | 186 (37.8%) | 148 (46.3%) | 38 (22.1%) | <0.0001 |
| | >1.2 | 306 (62.2%) | 172 (53.8%) | 134 (77.9%) | |
| Lacunar Syndrome | No | 450 (91.5%) | 280 (87.5%) | 170 (98.8%) | <0.0001 |
| | Yes | 42 (8.5%) | 40 (12.5%) | 2 (1.2%) | |
| History of Rankin score | 0 | 245 (49.8%) | 178 (55.6%) | 67 (39.0%) | 0.0014 |
| | 1 | 78 (15.9%) | 51 (15.9%) | 27 (15.7%) | |
| | 2 | 42 (8.5%) | 23 (7.2%) | 19 (11.0%) | |
| | >2 | 127 (25.8%) | 68 (21.3%) | 59 (34.3%) | |

| | | Total (N= 492) | Alive at 30 days (N= 320) | Deat at 30 days (N= 172) | P |
|---|---|---|---|---|---|
| Glasgow score | <5 | 53 (10.8%) | 4 (1.3%) | 49 (28.5%) | <0.0001 |
| | 5-12 | 113 (23.0%) | 44 (13.8%) | 69 (40.1%) | |
| | >12 | 326 (66.3%) | 272 (85.0%) | 54 (31.4%) | |
| NIHSS | <=5 | 162 (32.9%) | 145 (45.3%) | 17 (9.9%) | <0.0001 |
| | 6-15 | 149 (30.3%) | 107 (33.4%) | 42 (24.4%) | |
| | >15 | 181 (36.8%) | 68 (21.3%) | 113 (65.7%) | |
| Lobar hemorrhage | No | 297 (60.4%) | 180 (56.3%) | 117 (68.0%) | 0.0109 |
| | Yes | 195 (39.6%) | 140 (43.8%) | 55 (32.0%) | |
| Hemorrhage volume ($cm^3$) | < 30 | 305 (62.0%) | 245 (76.6%) | 60 (34.9%) | <0.0001 |
| | >=30 | 187 (38.0%) | 75 (23.4%) | 112 (65.1%) | |
| Prior antihypertensive medication | No | 246 (50.0%) | 169 (52.8%) | 77 (44.8%) | 0.0888 |
| | Yes | 246 (50.0%) | 151 (47.2%) | 95 (55.2%) | |
| Prior antiplatelet medication | No | 369 (75.0%) | 239 (74.7%) | 130 (75.6%) | 0.8272 |
| | Yes | 123 (25.0%) | 81 (25.3%) | 42 (24.4%) | |
| Prior anticoagulant medication | No | 391 (79.5%) | 272 (85.0%) | 119 (69.2%) | <0.0001 |
| | Yes | 101 (20.5%) | 48 (15.0%) | 53 (30.8%) | |
| Prior statine medication | No | 404 (82.1%) | 264 (82.5%) | 140 (81.4%) | 0.7605 |
| | Yes | 88 (17.9%) | 56 (17.5%) | 32 (18.6%) | |
| Prior fibrate medication | No | 472 (95.9%) | 308 (96.3%) | 164 (95.3%) | 0.6293 |
| | Yes | 20 (4.1%) | 12 (3.8%) | 8 (4.7%) | |
| Prior paracetamol medication | No | 365 (74.2%) | 247 (77.2%) | 118 (68.6%) | 0.0380 |
| | Yes | 127 (25.8%) | 73 (22.8%) | 54 (31.4%) | |

# C Descriptive statistics of all patients included before imputation

| | | Total (N= 738) | Alive at 30 days (N= 492) | Dead at 30 days (N= 246) | P |
|---|---|---|---|---|---|
| Age | N (N missing) | 1 (0.1%) | 1 (0.2%) | 0 (0.0%) | <0.0001 |
| | < 75 y | 320 (43.4%) | 249 (50.7%) | 71 (28.9%) | |
| | >=75 y | 417 (56.6%) | 242 (49.3%) | 175 (71.1%) | |
| Sex | N (N missing) | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0.1053 |
| | Male | 337 (45.7%) | 235 (47.8%) | 102 (41.5%) | |
| | Female | 401 (54.3%) | 257 (52.2%) | 144 (58.5%) | |
| Artheropatic diseases | Missing | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0.0031 |
| | No | 610 (82.7%) | 421 (85.6%) | 189 (76.8%) | |
| | Yes | 128 (17.3%) | 71 (14.4%) | 57 (23.2%) | |
| Cardiac arrhytmia | Missing | 17 (2.3%) | 13 (2.6%) | 4 (1.6%) | 0.0002 |
| | No | 557 (77.3%) | 390 (81.4%) | 167 (69.0%) | |
| | Yes | 164 (22.7%) | 89 (18.6%) | 75 (31.0%) | |
| Hypertension | Missing | 15 (2.0%) | 9 (1.8%) | 6 (2.4%) | 0.5001 |
| | No | 314 (43.4%) | 214 (44.3%) | 100 (41.7%) | |
| | Yes | 409 (56.6%) | 269 (55.7%) | 140 (58.3%) | |
| Diabetes | Missing | 20 (2.7%) | 10 (2.0%) | 10 (4.1%) | 0.8801 |
| | No | 634 (88.3%) | 425 (88.2%) | 209 (88.6%) | |
| | Yes | 84 (11.7%) | 57 (11.8%) | 27 (11.4%) | |
| Dyslipidemia | Missing | 41 (5.6%) | 20 (4.1%) | 21 (8.5%) | 0.8073 |
| | No | 506 (72.6%) | 344 (72.9%) | 162 (72.0%) | |
| | Yes | 191 (27.4%) | 128 (27.1%) | 63 (28.0%) | |
| Systolic blood pressure | Missing | 89 (12.1%) | 48 (9.8%) | 41 (16.7%) | 0.9304 |
| | Mean +/- SD | 169.23 +/- 32.68 | 169.77 +/- 30.93 | 168.04 +/- 36.23 | |
| Diastolic blood pressure | Missing | 92 (12.5%) | 51 (10.4%) | 41 (16.7%) | 0.7528 |
| | Mean +/- SD | 88.66 +/- 18.98 | 89.28 +/- 18.75 | 87.32 +/- 19.42 | |
| Body temperature | Missing | 214 (29.0%) | 124 (25.2%) | 90 (36.6%) | 0.0005 |
| | <=37.5 | 478 (91.2%) | 346 (94.0%) | 132 (84.6%) | |
| | >37.5 | 46 (8.8%) | 22 (6.0%) | 24 (15.4%) | |
| Heart rate | Missing | 98 (13.3%) | 61 (12.4%) | 37 (15.0%) | 0.2334 |
| | Mean +/- SD | 81.86 +/- 18.88 | 81.46 +/- 18.50 | 82.70 +/- 19.66 | |
| Oxygen saturation | Missing | 146 (19.8%) | 91 (18.5%) | 55 (22.4%) | 0.1345 |
| | Mean +/- SD | 96.58 +/- 3.28 | 96.57 +/- 2.87 | 96.60 +/- 4.03 | |

| | | Total (N= 738) | Alive at 30 days (N= 492) | Dead at 30 days (N= 246) | P |
|---|---|---|---|---|---|
| Blood glucose level (g/L) | Missing | 236 (32.0%) | 158 (32.1%) | 78 (31.7%) | <0.0001 |
| | <=1.2 | 197 (39.2%) | 159 (47.6%) | 38 (22.6%) | |
| | >1.2 | 305 (60.8%) | 175 (52.4%) | 130 (77.4%) | |
| Lacunar syndrome | Missing | 23 (3.1%) | 17 (3.5%) | 6 (2.4%) | 0.0001 |
| | Non | 649 (90.8%) | 417 (87.8%) | 232 (96.7%) | |
| | Oui | 66 (9.2%) | 58 (12.2%) | 8 (3.3%) | |
| History of Rankin score | Missing | 205 (27.7%) | 75 (15.2%) | 130 (52.9%) | <0.0001 |
| | 0 | 281 (52.7%) | 240 (57.6%) | 41 (35.3%) | |
| | 1 | 84 (15.8%) | 66 (15.8%) | 18 (15.5%) | |
| | 2 | 48 (9.0%) | 29 (7.0%) | 19 (16.4%) | |
| | >2 | 120 (22.5%) | 82 (19.7%) | 38 (32.8%) | |
| Glasgow score | Missing | 52 (7.0%) | 38 (7.7%) | 14 (5.7%) | <0.0001 |
| | <5 | 76 (11.1%) | 11 (2.4%) | 65 (28.0%) | |
| | 5-12 | 157 (22.9%) | 62 (13.7%) | 95 (40.9%) | |
| | >12 | 453 (66.0%) | 381 (83.9%) | 72 (31.0%) | |
| NIHSS | Missing | 118 (16.0%) | 55 (11.2%) | 63 (25.6%) | <0.0001 |
| | <=5 | 224 (36.1%) | 207 (47.4%) | 17 (9.3%) | |
| | 6-15 | 179 (28.9%) | 140 (32.0%) | 39 (21.3%) | |
| | >15 | 217 (35.0%) | 90 (20.6%) | 127 (69.4%) | |
| Lobar hemorrhage | Missing | 30 (4.1%) | 0 (0.0%) | 30 (12.2%) | 0.0314 |
| | No | 413 (58.3%) | 274 (55.7%) | 139 (64.4%) | |
| | Yes | 295 (41.7%) | 218 (44.3%) | 77 (35.6%) | |
| Hemorrhage volume ($cm^3$) | Missing | 65 (8.8%) | 20 (4.1%) | 45 (18.3%) | <0.0001 |
| | < 30 | 433 (64.3%) | 363 (76.9%) | 70 (34.8%) | |
| | >=30 | 240 (35.7%) | 109 (23.1%) | 131 (65.2%) | |
| Prior antihypertensive medication | Missing | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0.0328 |
| | No | 377 (51.1%) | 265 (53.9%) | 112 (45.5%) | |
| | Yes | 361 (48.9%) | 227 (46.1%) | 134 (54.5%) | |
| Prior antiplatelet medication | Missing | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0.2237 |
| | No | 560 (75.9%) | 380 (77.2%) | 180 (73.2%) | |
| | Yes | 178 (24.1%) | 112 (22.8%) | 66 (26.8%) | |
| Prior anticoagulant medication | Missing | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | <0.0001 |
| | No | 585 (79.3%) | 414 (84.1%) | 171 (69.5%) | |
| | Yes | 153 (20.7%) | 78 (15.9%) | 75 (30.5%) | |
| Prior statin medication | Missing | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0.8923 |
| | No | 605 (82.0%) | 404 (82.1%) | 201 (81.7%) | |
| | Yes | 133 (18.0%) | 88 (17.9%) | 45 (18.3%) | |

| | | Total (N= 738) | Alive at 30 days (N= 492) | Dead at 30 days (N= 246) | P |
|---|---|---|---|---|---|
| Prior fibrate medication | Missing | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0.2757 |
| | No | 710 (96.2%) | 476 (96.7%) | 234 (95.1%) | |
| | Yes | 28 (3.8%) | 16 (3.3%) | 12 (4.9%) | |
| Prior paracetamol medication | Missing | 0 (0.0%) | 0 (0.0%) | 0 (0.0%) | 0.0109 |
| | No | 558 (75.6%) | 386 (78.5%) | 172 (69.9%) | |
| | Yes | 180 (24.4%) | 106 (21.5%) | 74 (30.1%) | |

# D Descriptive statistics of all patients included after Predictive Mean Matching imputation

| | | Total (N= 738) | Alive at 30 days (N= 492) | Dead at 30 days (N= 246) | P |
|---|---|---|---|---|---|
| Age | < 75 y | 320 (43.4%) | 249 (50.6%) | 71 (28.9%) | <0.0001 |
| | >=75 y | 418 (56.6%) | 243 (49.4%) | 175 (71.1%) | |
| Sex | Male | 337 (45.7%) | 235 (47.8%) | 102 (41.5%) | 0.1053 |
| | Female | 401 (54.3%) | 257 (52.2%) | 144 (58.5%) | |
| Artheropatic diseases | No | 610 (82.7%) | 421 (85.6%) | 189 (76.8%) | 0.0031 |
| | Yes | 128 (17.3%) | 71 (14.4%) | 57 (23.2%) | |
| Cardiac arrhythmia | No | 570 (77.2%) | 400 (81.3%) | 170 (69.1%) | 0.0002 |
| | Yes | 168 (22.8%) | 92 (18.7%) | 76 (30.9%) | |
| Hypertension | No | 323 (43.8%) | 220 (44.7%) | 103 (41.9%) | 0.4626 |
| | Yes | 415 (56.2%) | 272 (55.3%) | 143 (58.1%) | |
| Diabetes | No | 653 (88.5%) | 435 (88.4%) | 218 (88.6%) | 0.9350 |
| | Yes | 85 (11.5%) | 57 (11.6%) | 28 (11.4%) | |
| Dyslipidemia | No | 540 (73.2%) | 362 (73.6%) | 178 (72.4%) | 0.7245 |
| | Yes | 198 (26.8%) | 130 (26.4%) | 68 (27.6%) | |
| Systolic blood pressure | N | 738 | 492 | 246 | 0.7299 |
| | Mean +/- SD | 168.90 +/- 32.79 | 169.08 +/- 31.09 | 168.53 +/- 36.01 | |
| Diastolic blood pressure | N | 738 | 492 | 246 | 0.7743 |
| | Mean +/- SD | 88.54 +/- 18.88 | 89.07 +/- 18.68 | 87.47 +/- 19.27 | |
| Body temperature (°C) | <=37.5 | 676 (91.6%) | 461 (93.7%) | 215 (87.4%) | 0.0036 |
| | >37.5 | 62 (8.4%) | 31 (6.3%) | 31 (12.6%) | |
| Heart rate | N | 738 | 492 | 246 | 0.1596 |
| | Mean +/- SD | 81.83 +/- 18.67 | 81.38 +/- 18.41 | 82.74 +/- 19.17 | |
| Oxygen saturation | N | 738 | 492 | 246 | 0.0867 |
| | Mean +/- SD | 96.62 +/- 3.11 | 96.59 +/- 2.79 | 96.70 +/- 3.67 | |
| Blood glucose level (g/L) | <=1.2 | 301 (40.8%) | 240 (48.8%) | 61 (24.8%) | <0.0001 |
| | >1.2 | 437 (59.2%) | 252 (51.2%) | 185 (75.2%) | |
| Lacunar syndrome | No | 666 (90.2%) | 428 (87.0%) | 238 (96.7%) | <0.0001 |
| | Yes | 72 (9.8%) | 64 (13.0%) | 8 (3.3%) | |
| History of Rankin score | 0 | 371 (50.3%) | 276 (56.1%) | 95 (38.6%) | <0.0001 |
| | 1 | 116 (15.7%) | 77 (15.7%) | 39 (15.9%) | |
| | 2 | 63 (8.5%) | 35 (7.1%) | 28 (11.4%) | |
| | >2 | 188 (25.5%) | 104 (21.1%) | 84 (34.1%) | |

| | | Total (N= 738) | Alive at 30 days (N= 492) | Dead at 30 days (N= 246) | P |
|---|---|---|---|---|---|
| Glasgow score | <5 | 82 (11.1%) | 11 (2.2%) | 71 (28.9%) | <0.0001 |
| | 5-12 | 166 (22.5%) | 67 (13.6%) | 99 (40.2%) | |
| | >12 | 490 (66.4%) | 414 (84.1%) | 76 (30.9%) | |
| NIHSS | <=5 | 259 (35.1%) | 235 (47.8%) | 24 (9.8%) | <0.0001 |
| | 6-15 | 213 (28.9%) | 158 (32.1%) | 55 (22.4%) | |
| | >15 | 266 (36.0%) | 99 (20.1%) | 167 (67.9%) | |
| Lobar hemorrhage | No | 437 (59.2%) | 274 (55.7%) | 163 (66.3%) | 0.0059 |
| | Yes | 301 (40.8%) | 218 (44.3%) | 83 (33.7%) | |
| Hemorrhage volume ($cm^3$) | < 30 | 466 (63.1%) | 379 (77.0%) | 87 (35.4%) | <0.0001 |
| | >=30 | 272 (36.9%) | 113 (23.0%) | 159 (64.6%) | |
| Prior antihypertensive medication | No | 377 (51.1%) | 265 (53.9%) | 112 (45.5%) | 0.0328 |
| | Yes | 361 (48.9%) | 227 (46.1%) | 134 (54.5%) | |
| Prior antiplatelet medication | No | 560 (75.9%) | 380 (77.2%) | 180 (73.2%) | 0.2237 |
| | Yes | 178 (24.1%) | 112 (22.8%) | 66 (26.8%) | |
| Prior anticoagulant medication | No | 585 (79.3%) | 414 (84.1%) | 171 (69.5%) | <0.0001 |
| | Yes | 153 (20.7%) | 78 (15.9%) | 75 (30.5%) | |
| Prior statin medication | No | 605 (82.0%) | 404 (82.1%) | 201 (81.7%) | 0.8923 |
| | Yes | 133 (18.0%) | 88 (17.9%) | 45 (18.3%) | |
| Prior fibrate medication | No | 710 (96.2%) | 476 (96.7%) | 234 (95.1%) | 0.2757 |
| | Yes | 28 (3.8%) | 16 (3.3%) | 12 (4.9%) | |
| Prior paracetamol medication | No | 558 (75.6%) | 386 (78.5%) | 172 (69.9%) | 0.0109 |
| | Yes | 180 (24.4%) | 106 (21.5%) | 74 (30.1%) | |

# E Descriptive statistics of all patients included in complete case-analysis

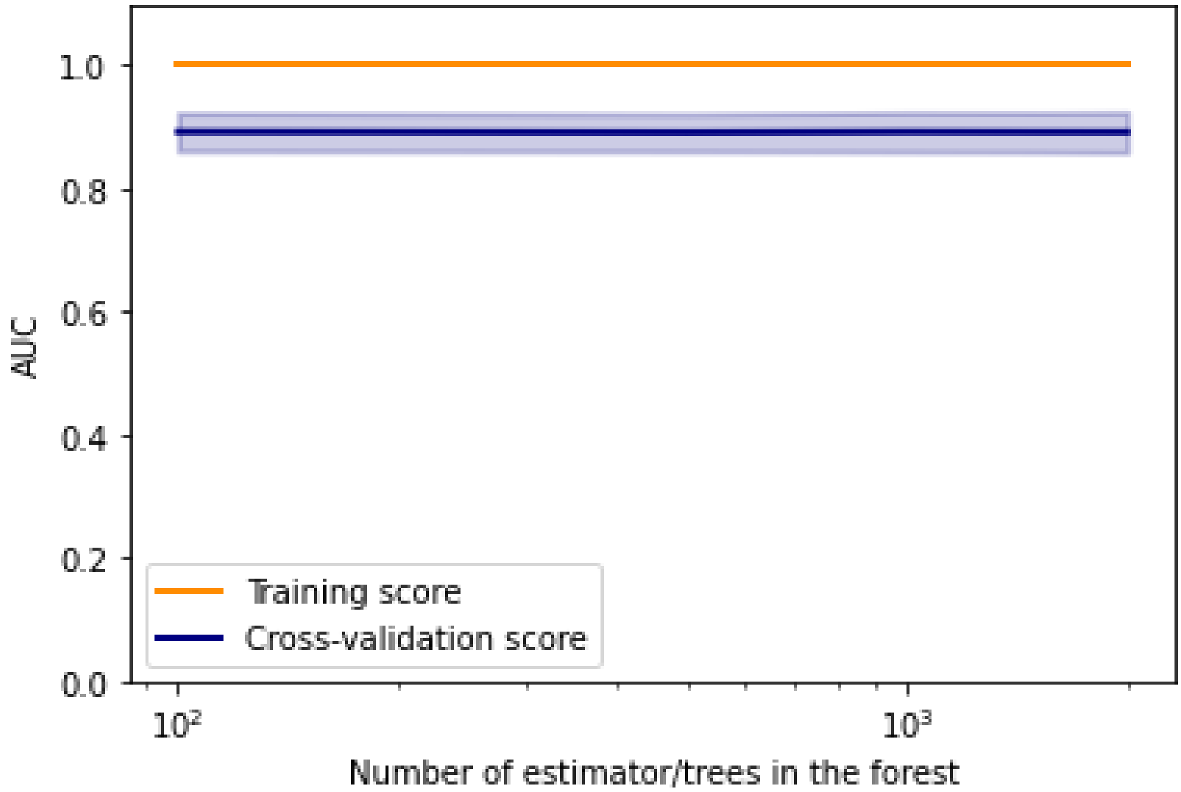
| | | Total (N= 258) | Alive at 30 days (N= 223) | Dead at 30 days (N= 35) | P |
|---|---|---|---|---|---|
| Age | < 75 y | 118 (45.7%) | 112 (50.2%) | 6 (17.1%) | 0.0003 |
| | >=75 y | 140 (54.3%) | 111 (49.8%) | 29 (82.9%) | |
| Sex | Male | 127 (49.2%) | 109 (48.9%) | 18 (51.4%) | 0.7791 |
| | Female | 131 (50.8%) | 114 (51.1%) | 17 (48.6%) | |
| Artheropatic diseases | No | 221 (85.7%) | 195 (87.4%) | 26 (74.3%) | 0.0389 |
| | Yes | 37 (14.3%) | 28 (12.6%) | 9 (25.7%) | |
| Cardiac arrhythmia | No | 196 (76.0%) | 175 (78.5%) | 21 (60.0%) | 0.0174 |
| | Yes | 62 (24.0%) | 48 (21.5%) | 14 (40.0%) | |
| Hypertension | No | 111 (43.0%) | 99 (44.4%) | 12 (34.3%) | 0.2614 |
| | Yes | 147 (57.0%) | 124 (55.6%) | 23 (65.7%) | |
| Diabetes | No | 222 (86.0%) | 193 (86.5%) | 29 (82.9%) | 0.5993 |
| | Yes | 36 (14.0%) | 30 (13.5%) | 6 (17.1%) | |
| Dyslipidemia | No | 182 (70.5%) | 159 (71.3%) | 23 (65.7%) | 0.5003 |
| | Yes | 76 (29.5%) | 64 (28.7%) | 12 (34.3%) | |
| Pression artérielle systolique | N | 258 | 223 | 35 | 0.4737 |
| | Mean +/- SD | 170.74 +/- 31.38 | 170.48 +/- 31.59 | 172.37 +/- 30.40 | |
| Diastolic blood pressure | N | 258 | 223 | 35 | 0.9041 |
| | Mean +/- SD | 89.72 +/- 17.72 | 89.96 +/- 17.99 | 88.20 +/- 16.05 | |
| Body temperature (°C) | <=37.5 | 237 (91.9%) | 210 (94.2%) | 27 (77.1%) | 0.0029 |
| | >37.5 | 21 (8.1%) | 13 (5.8%) | 8 (22.9%) | |
| Heart rate | N | 258 | 223 | 35 | 0.1456 |
| | Mean +/- SD | 81.35 +/- 17.02 | 80.72 +/- 16.75 | 85.40 +/- 18.42 | |
| Oxygen saturation | N | 258 | 223 | 35 | 0.7861 |
| | Mean +/- SD | 96.44 +/- 3.07 | 96.47 +/- 3.06 | 96.26 +/- 3.18 | |
| Blood glucose level (g/L) | <=1.2 | 118 (45.7%) | 108 (48.4%) | 10 (28.6%) | 0.0283 |
| | >1.2 | 140 (54.3%) | 115 (51.6%) | 25 (71.4%) | |
| Lacunar syndrome | No | 226 (87.6%) | 193 (86.5%) | 33 (94.3%) | 0.2734 |
| | Yes | 32 (12.4%) | 30 (13.5%) | 2 (5.7%) | |
| History of Rankin score | 0 | 135 (52.3%) | 126 (56.5%) | 9 (25.7%) | 0.0045 |
| | 1 | 55 (21.3%) | 44 (19.7%) | 11 (31.4%) | |
| | 2 | 18 (7.0%) | 14 (6.3%) | 4 (11.4%) | |
| | >2 | 50 (19.4%) | 39 (17.5%) | 11 (31.4%) | |

| | | | | | |
|---|---|---|---|---|---|
| Glasgow score | <5 | 8 (3.1%) | 3 (1.3%) | 5 (14.3%) | <0.0001 |
| | 5-12 | 44 (17.1%) | 30 (13.5%) | 14 (40.0%) | |
| | >12 | 206 (79.8%) | 190 (85.2%) | 16 (45.7%) | |
| NIHSS | <=5 | 114 (44.2%) | 110 (49.3%) | 4 (11.4%) | <0.0001 |
| | 6-15 | 79 (30.6%) | 66 (29.6%) | 13 (37.1%) | |
| | >15 | 65 (25.2%) | 47 (21.1%) | 18 (51.4%) | |
| Lobar hemorrhage | No | 142 (55.0%) | 123 (55.2%) | 19 (54.3%) | 0.9233 |
| | Yes | 116 (45.0%) | 100 (44.8%) | 16 (45.7%) | |
| Hemorrhage volume ($cm^3$) | < 30 | 186 (72.1%) | 170 (76.2%) | 16 (45.7%) | 0.0002 |
| | >=30 | 72 (27.9%) | 53 (23.8%) | 19 (54.3%) | |
| Prior antihypertensive medication | No | 138 (53.5%) | 120 (53.8%) | 18 (51.4%) | 0.7927 |
| | Yes | 120 (46.5%) | 103 (46.2%) | 17 (48.6%) | |
| Prior antiplatelet medication | No | 198 (76.7%) | 174 (78.0%) | 24 (68.6%) | 0.2183 |
| | Yes | 60 (23.3%) | 49 (22.0%) | 11 (31.4%) | |
| Prior anticoagulant medication | No | 217 (84.1%) | 189 (84.8%) | 28 (80.0%) | 0.4745 |
| | Yes | 41 (15.9%) | 34 (15.2%) | 7 (20.0%) | |
| Prior statin medication | No | 207 (80.2%) | 179 (80.3%) | 28 (80.0%) | 0.9704 |
| | Yes | 51 (19.8%) | 44 (19.7%) | 7 (20.0%) | |
| Prior fibrate medication | No | 249 (96.5%) | 217 (97.3%) | 32 (91.4%) | 0.1081 |
| | Yes | 9 (3.5%) | 6 (2.7%) | 3 (8.6%) | |
| Prior paracetamol medication | No | 196 (76.0%) | 171 (76.7%) | 25 (71.4%) | 0.4989 |
| | Yes | 62 (24.0%) | 52 (23.3%) | 10 (28.6%) | |

# F Parameters tested and selected on the training datasets

| | **Tested parameters** | **Best parameters in complete dataset** | **Best parameters in imputed dataset** |
|---|---|---|---|
| *n_estimators* | 100, 500, 1000, 1500, 2000 | 100 | 100 |
| *max_features* | sqrt, log2, None | None | log2 |
| *min_samples_split* | From 2 to 100 | 47 | 10 |
| *max_depth* | 3, 6, 9, 12, 15, 20, Nones | None | None |

## G Validation curve of random forest depending on the number of trees



# H Descriptive statistics of complete case-analysis testing dataset

| | | Total (N= 86) | Alive at 30 days (N= 77) | Dead at 30 days (N= 9) | P |
|---|---|---|---|---|---|
| Age | >=75 y | 47 (54.7%) | 38 (49.4%) | 9 (100.0%) | 0.0034 |
| | < 75 y | 39 (45.3%) | 39 (50.6%) | 0 (0.0%) | |
| Sex | Male | 41 (47.7%) | 38 (49.4%) | 3 (33.3%) | 0.4881 |
| | Female | 45 (52.3%) | 39 (50.6%) | 6 (66.7%) | |
| Artheromatic diseases | No | 72 (83.7%) | 66 (85.7%) | 6 (66.7%) | 0.1590 |
| | Yes | 14 (16.3%) | 11 (14.3%) | 3 (33.3%) | |
| Cardiac arrhythmia | No | 69 (80.2%) | 63 (81.8%) | 6 (66.7%) | 0.3718 |
| | Yes | 17 (19.8%) | 14 (18.2%) | 3 (33.3%) | |
| Hypertension | No | 33 (38.4%) | 32 (41.6%) | 1 (11.1%) | 0.1441 |
| | Yes | 53 (61.6%) | 45 (58.4%) | 8 (88.9%) | |
| Diabetes | No | 77 (89.5%) | 70 (90.9%) | 7 (77.8%) | 0.2374 |
| | Yes | 9 (10.5%) | 7 (9.1%) | 2 (22.2%) | |
| Dyslipidemia | No | 55 (64.0%) | 50 (64.9%) | 5 (55.6%) | 0.7166 |
| | Yes | 31 (36.0%) | 27 (35.1%) | 4 (44.4%) | |
| Systolic blood pressure | N | 86 | 77 | 9 | 0.3125 |
| | Mean +/- SD | 169.37 +/- 30.42 | 168.49 +/- 31.25 | 176.89 +/- 22.08 | |
| Diastolic blood pressure | N | 86 | 77 | 9 | 0.7093 |
| | Mean +/- SD | 91.30 +/- 16.69 | 91.29 +/- 17.37 | 91.44 +/- 9.55 | |
| Body temperature (°C) | >37.5 | 5 (5.8%) | 2 (2.6%) | 3 (33.3%) | 0.0073 |
| | <=37.5 | 81 (94.2%) | 75 (97.4%) | 6 (66.7%) | |
| Heart rate | N | 86 | 77 | 9 | 0.0649 |
| | Mean +/- SD | 82.14 +/- 19.04 | 80.90 +/- 18.62 | 92.78 +/- 20.45 | |
| Oxygen saturation (%) | N | 86 | 77 | 9 | 0.5269 |
| | Mean +/- SD | 96.69 +/- 2.26 | 96.77 +/- 2.14 | 96.00 +/- 3.20 | |
| Blood glucose level (g/L) | >1.2 | 52 (60.5%) | 46 (59.7%) | 6 (66.7%) | 1.0000 |
| | <=1.2 | 34 (39.5%) | 31 (40.3%) | 3 (33.3%) | |
| Lacunar syndrome | No | 73 (84.9%) | 65 (84.4%) | 8 (88.9%) | 1.0000 |
| | Yes | 13 (15.1%) | 12 (15.6%) | 1 (11.1%) | |
| History of Rankin score | >2 | 15 (17.4%) | 11 (14.3%) | 4 (44.4%) | 0.1036 |
| | 0 | 52 (60.5%) | 49 (63.6%) | 3 (33.3%) | |
| | 1 | 15 (17.4%) | 13 (16.9%) | 2 (22.2%) | |
| | 2 | 4 (4.7%) | 4 (5.2%) | 0 (0.0%) | |

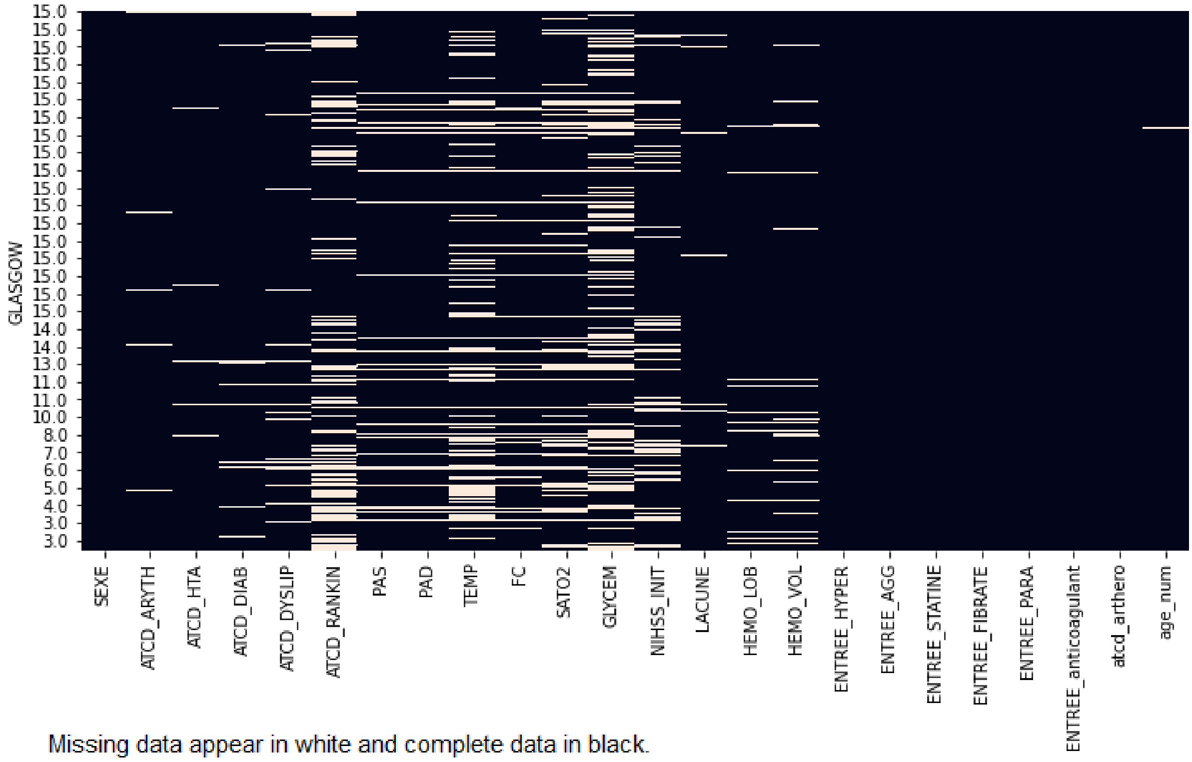
| | | Total (N= 86) | Alive at 30 days (N= 77) | Dead at 30 days (N= 9) | P |
|---|---|---|---|---|---|
| Glasgow score | 5-12 | 13 (15.1%) | 11 (14.3%) | 2 (22.2%) | 0.0165 |
| | >12 | 70 (81.4%) | 65 (84.4%) | 5 (55.6%) | |
| | <5 | 3 (3.5%) | 1 (1.3%) | 2 (22.2%) | |
| NIHSS | 6-15 | 28 (32.6%) | 26 (33.8%) | 2 (22.2%) | 0.0168 |
| | >15 | 22 (25.6%) | 16 (20.8%) | 6 (66.7%) | |
| | <=5 | 36 (41.9%) | 35 (45.5%) | 1 (11.1%) | |
| Lobar hemorrhage | No | 46 (53.5%) | 41 (53.2%) | 5 (55.6%) | 1.0000 |
| | Yes | 40 (46.5%) | 36 (46.8%) | 4 (44.4%) | |
| Hemorrhage volume ($cm^3$) | >=30 | 21 (24.4%) | 17 (22.1%) | 4 (44.4%) | 0.2127 |
| | < 30 | 65 (75.6%) | 60 (77.9%) | 5 (55.6%) | |
| Prior antihypertensive. medication | No | 44 (51.2%) | 40 (51.9%) | 4 (44.4%) | 0.7357 |
| | Yes | 42 (48.8%) | 37 (48.1%) | 5 (55.6%) | |
| Prior antiplatelet medication | No | 69 (80.2%) | 63 (81.8%) | 6 (66.7%) | 0.3718 |
| | Yes | 17 (19.8%) | 14 (18.2%) | 3 (33.3%) | |
| Prior anticoagulation medication | No | 76 (88.4%) | 68 (88.3%) | 8 (88.9%) | 1.0000 |
| | Yes | 10 (11.6%) | 9 (11.7%) | 1 (11.1%) | |
| Prior statine medication | No | 62 (72.1%) | 55 (71.4%) | 7 (77.8%) | 1.0000 |
| | Yes | 24 (27.9%) | 22 (28.6%) | 2 (22.2%) | |
| Prior fibrate medication | No | 83 (96.5%) | 75 (97.4%) | 8 (88.9%) | 0.2852 |
| | Yes | 3 (3.5%) | 2 (2.6%) | 1 (11.1%) | |
| Prior paracetamol medication | No | 69 (80.2%) | 63 (81.8%) | 6 (66.7%) | 0.3718 |
| | Yes | 17 (19.8%) | 14 (18.2%) | 3 (33.3%) | |

# I Descriptive statistics of imputed testing dataset

| | | Total (N= 246) | Alive at 30 days (N= 172) | Dead at 30 days (N= 74) | P |
|---|---|---|---|---|---|
| Age | < 75 y | 111 (45.1%) | 88 (51.2%) | 23 (31.1%) | 0.0037 |
| | >=75 y | 135 (54.9%) | 84 (48.8%) | 51 (68.9%) | |
| Sex | Male | 115 (46.7%) | 85 (49.4%) | 30 (40.5%) | 0.2006 |
| | Female | 131 (53.3%) | 87 (50.6%) | 44 (59.5%) | |
| Artheropatic diseases | No | 215 (87.4%) | 151 (87.8%) | 64 (86.5%) | 0.7774 |
| | Yes | 31 (12.6%) | 21 (12.2%) | 10 (13.5%) | |
| Cardiac arrhythmia | No | 190 (77.2%) | 137 (79.7%) | 53 (71.6%) | 0.1684 |
| | Yes | 56 (22.8%) | 35 (20.3%) | 21 (28.4%) | |
| Hypertension | No | 115 (46.7%) | 82 (47.7%) | 33 (44.6%) | 0.6570 |
| | Yes | 131 (53.3%) | 90 (52.3%) | 41 (55.4%) | |
| Diabetes | No | 212 (86.2%) | 151 (87.8%) | 61 (82.4%) | 0.2641 |
| | Yes | 34 (13.8%) | 21 (12.2%) | 13 (17.6%) | |
| Dyslipidemia | No | 174 (70.7%) | 125 (72.7%) | 49 (66.2%) | 0.3073 |
| | Yes | 72 (29.3%) | 47 (27.3%) | 25 (33.8%) | |
| Systolic blood pressure | N | 246 | 172 | 74 | 0.1295 |
| | Mean +/- SD | 167.69 +/- 30.34 | 165.53 +/- 29.74 | 172.69 +/- 31.31 | |
| Diastolic blood pressure | N | 246 | 172 | 74 | 0.9634 |
| | Mean +/- SD | 87.12 +/- 18.94 | 87.16 +/- 17.86 | 87.01 +/- 21.35 | |
| Body temperature (°C) | <=37.5 | 221 (89.8%) | 159 (92.4%) | 62 (83.8%) | 0.0393 |
| | >37.5 | 25 (10.2%) | 13 (7.6%) | 12 (16.2%) | |
| Heart rate | N | 246 | 172 | 74 | 0.0267 |
| | Mean +/- SD | 81.43 +/- 19.53 | 80.35 +/- 20.37 | 83.95 +/- 17.30 | |
| Oxygen saturation | N | 246 | 172 | 74 | 0.4509 |
| | Mean +/- SD | 96.53 +/- 3.13 | 96.38 +/- 3.43 | 96.89 +/- 2.30 | |
| Blood glucose level (g/L) | <=1.2 | 115 (46.7%) | 92 (53.5%) | 23 (31.1%) | 0.0012 |
| | >1.2 | 131 (53.3%) | 80 (46.5%) | 51 (68.9%) | |
| Lacunar syndrome | No | 216 (87.8%) | 148 (86.0%) | 68 (91.9%) | 0.1988 |
| | Yes | 30 (12.2%) | 24 (14.0%) | 6 (8.1%) | |
| History of Rankin score | 0 | 126 (51.2%) | 98 (57.0%) | 28 (37.8%) | 0.0318 |
| | 1 | 38 (15.4%) | 26 (15.1%) | 12 (16.2%) | |
| | 2 | 21 (8.5%) | 12 (7.0%) | 9 (12.2%) | |
| | >2 | 61 (24.8%) | 36 (20.9%) | 25 (33.8%) | |

| | | Total (N= 246) | Alive at 30 days (N= 172) | Dead at 30 days (N= 74) | P |
|---|---|---|---|---|---|
| Glasgow score | <5 | 29 (11.8%) | 7 (4.1%) | 22 (29.7%) | <0.0001 |
| | 5-12 | 53 (21.5%) | 23 (13.4%) | 30 (40.5%) | |
| | >12 | 164 (66.7%) | 142 (82.6%) | 22 (29.7%) | |
| NIHSS | <=5 | 97 (39.4%) | 90 (52.3%) | 7 (9.5%) | <0.0001 |
| | 6-15 | 64 (26.0%) | 51 (29.7%) | 13 (17.6%) | |
| | >15 | 85 (34.6%) | 31 (18.0%) | 54 (73.0%) | |
| Lobar hemorrhage | No | 140 (56.9%) | 94 (54.7%) | 46 (62.2%) | 0.2753 |
| | Yes | 106 (43.1%) | 78 (45.3%) | 28 (37.8%) | |
| Hemorrhage volume ($cm^3$) | < 30 | 161 (65.4%) | 134 (77.9%) | 27 (36.5%) | <0.0001 |
| | >=30 | 85 (34.6%) | 38 (22.1%) | 47 (63.5%) | |
| Prior antihypertensive medication | No | 131 (53.3%) | 96 (55.8%) | 35 (47.3%) | 0.2195 |
| | Yes | 115 (46.7%) | 76 (44.2%) | 39 (52.7%) | |
| Prior antiplatelet medication | No | 191 (77.6%) | 141 (82.0%) | 50 (67.6%) | 0.0129 |
| | Yes | 55 (22.4%) | 31 (18.0%) | 24 (32.4%) | |
| Prior anticoagulant medication | No | 194 (78.9%) | 142 (82.6%) | 52 (70.3%) | 0.0304 |
| | Yes | 52 (21.1%) | 30 (17.4%) | 22 (29.7%) | |
| Prior statin medication | No | 201 (81.7%) | 140 (81.4%) | 61 (82.4%) | 0.8470 |
| | Yes | 45 (18.3%) | 32 (18.6%) | 13 (17.6%) | |
| Prior fibrate medication | No | 238 (96.7%) | 168 (97.7%) | 70 (94.6%) | 0.2465 |
| | Yes | 8 (3.3%) | 4 (2.3%) | 4 (5.4%) | |
| Priori paracetamol medication | No | 193 (78.5%) | 139 (80.8%) | 54 (73.0%) | 0.1701 |
| | Yes | 53 (21.5%) | 33 (19.2%) | 20 (27.0%) | |

## J Heatmap of missing data based on Glasgow score




Missing data appear in white and complete data in black.

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# Serment d'Hippocrate

Au moment d'être admis à exercer la médecine, je promets et je jure d'être fidèle aux lois de l'honneur et de la probité.

Mon premier souci sera de rétablir, de préserver ou de promouvoir la santé dans tous ses éléments, physiques et mentaux, individuels et sociaux.

Je respecterai toutes les personnes, leur autonomie et leur volonté, sans aucune discrimination selon leur état ou leurs convictions. J'interviendrai pour les protéger si elles sont affaiblies, vulnérables ou menacées dans leur intégrité ou leur dignité. Même sous la contrainte, je ne ferai pas usage de mes connaissances contre les lois de l'humanité.

J'informerai les patients des décisions envisagées, de leurs raisons et de leurs conséquences. Je ne tromperai jamais leur confiance et n'exploiterai pas le pouvoir hérité des circonstances pour forcer les consciences.

Je donnerai mes soins à l'indigent et à quiconque me le demandera. Je ne me laisserai pas influencer par la soif du gain ou la recherche de la gloire.

Admis dans l'intimité des personnes, je tairai les secrets qui me seront confiés. Reçu à l'intérieur des maisons, je respecterai les secrets des foyers et ma conduite ne servira pas à corrompre les mœurs.

Je ferai tout pour soulager les souffrances. Je ne prolongerai pas abusivement les agonies. Je ne provoquerai jamais la mort délibérément.

Je préserverai l'indépendance nécessaire à l'accomplissement de ma mission. Je n'entreprendrai rien qui dépasse mes compétences. Je les entretiendrai et les perfectionnerai pour assurer au mieux les services qui me seront demandés.

J'apporterai mon aide à mes confrères ainsi qu'à leurs familles dans l'adversité.

Que les hommes et mes confrères m'accordent leur estime si je suis fidèle à mes promesses ; que je sois déshonoré et méprisé si j'y manque.

FACULTÉ
DE MÉDECINE &
SCIENCES DE LA SANTÉ

# THESE DE DOCTORAT EN MEDECINE

**INTERNE**

Madame / Monsieur **QUERE Victor**
Inscrit-e en DES de **Neurologie**

Titre définitif de la thèse soutenue :

**Comparison of logistic regression analysis with random forest for prediction of ICH early mortality in the population-based Brest Stroke Registry**

**ACCORD DU PRESIDENT DU JURY DE THESE SUR L'IMPRESSION DE LA THESE**

**OUI :** ☒
**NON :** ☐

La présente autorisation d'imprimer sa thèse est délivrée à l'interne susmentionné.

Brest, le 28/09/21

**Le Président du jury de thèse**

S. Timsit

**La Directrice de l'UFR de médecine et des sciences de la santé de Brest**

B. COCHENER

UNIVERSITÉ DE BRETAGNE OCCIDENTALE
FACULTÉ DE MÉDECINE DE BREST

QUERE (Victor) – Comparison of logistic regression analysis with random forest for prediction of ICH early mortality in the population-based Brest Stroke Registry- 54 f. , 5 ill ., 10 tabl.
Th. : Méd. : Brest 2021

**RESUME** : Après avoir rappelé l'importance de l'établissement d'un pronostic à la phase aigüe de l'hémorragie intracérébrale spontanée, l'auteur décrit et compare quatre scores pronostics.

Deux d'entre eux sont issus d'une méthode de régression et les deux autres sont basés sur un algorithme de forêts aléatoires. Ces scores sont créés respectivement sur une base de données imputée et une base de données de laquelle ont été exclues les observations à données manquantes.

Après avoir comparé les performances des outils ainsi créés, il discute le poids des variables utilisées pour définir le pronostic. Par ailleurs, il aborde la problématique liée aux données manquantes et leur possible influence sur les résultats obtenus.

Il conclut enfin en discutant les différents intérêts des outils d'intelligence artificielle dans l'établissement du pronostic des hémorragie intracérébrales spontanées.

**MOTS CLES** :
HEMORRAGIE INTRACEREBRALE SPONTANEE
PRONOSTIC
INTELLIGENCE ARTIFICIELLE
DONNEES MANQUANTES
FORETS ALEATOIRES

**JURY** :

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